Universal Optimality of the Planar Triangular Lattice

Conditional Corollaries (D1–D4) with Complete Pen-and-Paper Proofs,
and a *Math Sup / Math Spé* Foundational Reference Appendix

Prepared for Print & Pen-and-Paper Peer Review
Based on the `openai/math` release (AI-generated, not peer reviewed)
& the *SocrateAI-Scientific-Elenchus* Textbook

October 2026

Abstract

This document is a self-contained, human-readable exposition of the universal optimality statement for the two-dimensional triangular (hexagonal) lattice $A \subset \mathbb{R}^2$: A minimizes the lower pair-interaction energy per particle among all locally finite planar configurations of density one, simultaneously for every completely monotone potential of squared distance. The statement is *not* proved here in full. Its entire difficulty—the existence, for every $\alpha > 0$, of a sharp two-dimensional auxiliary function for the Gaussian G_α—is imported from the `openai/math` release as Imported Claim 2.6, an AI-generated and not yet peer-reviewed result; as of November 2025 the conjecture is still described as open in the literature [Leb25]. Everything downstream of that claim is proved completely: the Cohn–Kumar linear-programming bound for periodic configurations (Section 2), the passage from Gaussians to all completely monotone potentials via Bernstein, Fatou and Tonelli (Section 3), and the four extensions (Section 4): Riesz potentials and the Epstein zeta threshold (D1 & D1b), rescaling to arbitrary densities (D2), the vacancy theorem and non-uniqueness of minimizers (D3), and the equivalence of open and closed disk boundaries (D4). Part V is a foundational appendix written for readers trained in the French *Math Sup / Math Spé* (*Classes Préparatoires*) curriculum, bridging classical linear algebra and real analysis to measure theory, two-dimensional Poisson summation, and the double-zero structure of auxiliary functions.

Epistemic status (read this first).

- **Proved here, unconditionally:** Propositions 1.5 and 1.7; Theorem 2.3 (LP bound, periodic case); Proposition 2.4; Lemma 3.1; the derivation of Theorem 1.10 from Imported Claim 2.6 (Section 3); Results D1–D4 as consequences of Theorem 1.10 (Section 4); every statement in the Appendix.
- **Standard, cited but not reproduced:** Bernstein's representation theorem (Theorem 3.2, [Ber29, Wid41]); the linear-programming bound for *arbitrary* locally finite configurations (Proposition 2.5, [CdCI18, Prop. 2.2]).
- **Imported and *not* verified in this document:** Imported Claim 2.6—the existence, for every $\alpha > 0$, of a sharp 2D auxiliary function (Definition 2.1). Its source is the theorem `OAI.AtomicTriangular.universal_energy_minimum` of the `openai/math` release [OAI26], which is AI-generated. The Lean kernel accepts that proof against the release's own definitions, but the fidelity of those definitions to the Cohn–Kumar/CKMRV notions and the accompanying manuscript have not been peer reviewed [XC26]. Among *lattices*, the result is Montgomery's theorem [Mon88]; among all configurations it is, in the peer-reviewed literature, an open conjecture [Leb25].
- **Consequently**, Theorem 1.10 and Results D1–D4 are *conditional* on Imported Claim 2.6. Nothing in this document should be cited as a proof of the two-dimensional universal optimality conjecture.

Contents

1 The Mathematical Setup and Definitions **3**
- 1.1 Point Configurations, Local Finiteness, and Centred Density . . . 3
- 1.2 The Triangular Lattice A and Its Dual Lattice A^* . . . 3
- 1.3 Admissible Potentials and Lower Energy . . . 5

2 The Linear Programming Bound and the 2D Auxiliary Function **5**
- 2.1 The Sharp Auxiliary ("Magic") Function Conditions . . . 6
- 2.2 Complete Proof of the LP Bound for Periodic Configurations . . . 6
- 2.3 Connecting Periodic Energy to Lower Disk Energy . . . 7
- 2.4 What Remains: Non-Periodic Competitors and the 2D Auxiliary Function (Not Proved Here) . . . 8

3 From Gaussians to All Completely Monotone Potentials **9**
- 3.1 The Constant Potential Lemma (`energy_one`) . . . 9
- 3.2 Proof of Theorem 1.10 via Bernstein's Theorem, Fatou, and Tonelli . . . 10

4 Complete Proofs of the Four Corollaries (Results D1–D4) **11**
- 4.1 Result D1: Riesz Potentials ($V(r) = r^{-s}$) . . . 11
- 4.2 Result D1b: Finiteness of the Riesz Lattice Energy for $s > 2$ and Divergence for $s \leq 2$ 12
- 4.3 Result D2: Universal Optimality at Every Density $\rho > 0$ . . . 13
- 4.4 Result D3: The Vacancy Theorem and Non-Uniqueness of Minimizers . . . 14
- 4.5 Result D4: Equivalence of Open and Closed Disks . . . 15

Part V: Foundational Reference Appendix (Math Sup / Math Spé Refresher) **17**

A Euclidean Spaces, Gram Matrices, and Dual Lattices **17**
- A.1 Covolume and Gram Matrix . . . 17
- A.2 Dual Basis and Dual Lattice Λ^* . . . 17
- A.3 Asymptotic Lattice Point Counting . . . 18

B Limit Inferior (lim inf), Limit Superior (lim sup), and $[0, \infty]$ **18**
- B.1 Definition Over a Continuous Parameter $R \to \infty$ . . . 18
- B.2 Why lim inf Is Super-Additive (and Why Equality Fails) . . . 18

C From *Math Spé* Riemann Integrals to Measure Theory, Fatou, and Tonelli **19**
- C.1 Borel Measures μ on $[0, \infty)$: Unifying Sums and Densities . . . 19
- C.2 Fatou's Lemma and Tonelli's Theorem . . . 19

D Completely Monotone Functions, Leibniz's Rule, and Gamma Integrals **19**
- D.1 Closure Properties of Completely Monotone Functions . . . 19

E 2D Fourier Analysis, Gaussian Self-Duality, and Poisson Summation **20**
- E.1 Fourier Transform of a Gaussian via a *Math Spé* Differential Equation . . . 20
- E.2 The 2D Poisson Summation Formula from *Math Spé* Fourier Series . . . 21

F Double Zeros and Sine-Product Cardinal Interpolation **21**
- F.1 Why Minorants Must Have Double Zeros at Contact Points . . . 21
- F.2 From Euler's Sine Product to Cardinal Interpolation with Double Zeros . . . 21

G Printable Pen-and-Paper Review Checklist **22**

1 The Mathematical Setup and Definitions

We work in the Euclidean plane $\mathbb{R}^2$ equipped with the standard scalar product $\langle x, y\rangle = x_1y_1+x_2y_2$ and Euclidean norm $\|x\| = \sqrt{x_1^2 + x_2^2}$. For any radius $R \in \mathbb{R}$, we denote by

$$\overline{B}(0, R) = \{x \in \mathbb{R}^2 : \|x\| \le R\} \quad \text{and} \quad B^\circ(0, R) = \{x \in \mathbb{R}^2 : \|x\| < R\}$$

the closed and open disks of radius R centered at the origin.

1.1 Point Configurations, Local Finiteness, and Centred Density

A **point configuration** is an arbitrary subset $C \subseteq \mathbb{R}^2$. Because C is a set without multiplicity, no two particles occupy the exact same position.

Definition 1.1 (Local finiteness and disk counts)**.** A configuration $C \subseteq \mathbb{R}^2$ is **locally finite** if the intersection

$$C_R := C \cap \overline{B}(0, R)$$

is a finite set for every $R \in \mathbb{R}$. When C is locally finite, we denote by $N_R(C) = \text{card}(C_R) \in \mathbb{N}$ the number of points of C inside the closed disk $\overline{B}(0, R)$.

Definition 1.2 (Centred density)**.** A locally finite configuration $C \subseteq \mathbb{R}^2$ has **centred density** ρ (for $\rho > 0$) if

$$\lim_{R\to\infty} \frac{N_R(C)}{\pi R^2} = \rho.$$

In particular, C has **density one** if $\lim_{R\to\infty} \frac{N_R(C)}{\pi R^2} = 1$.

Remark 1.3 (Statement-adequacy notes for reviewers)*.* Notice two features of Definition 1.2:

(i) *Centred vs. uniform density:* The limit is taken over disks centered at the origin 0. For any other fixed center $c \in \mathbb{R}^2$, the triangle inequality gives $\overline{B}(0, R - \|c\|) \subseteq \overline{B}(c, R) \subseteq \overline{B}(0, R + \|c\|)$, which implies $\lim_{R\to\infty} \text{card}(C \cap \overline{B}(c, R))/(\pi R^2) = \rho$ as well. However, centred density does not prevent sparse, localized clusters far from the origin (e.g., placing k^2 points in a unit disk around $(2^k, 0)$ for each $k \ge 1$ adds only $O((\log R)^3)$ points inside radius R, leaving the asymptotic density unchanged).

(ii) *Closed vs. open disks:* Counting points in $\overline{B}(0, R)$ vs. $B^\circ(0, R)$ can differ on the boundary circle $\|x\| = R$. In Section 4.5, we prove (Result D4) that for locally finite configurations, open and closed disks yield the exact same asymptotic density and lower energy.

1.2 The Triangular Lattice A and Its Dual Lattice A^*

A lattice $\Lambda \subset \mathbb{R}^2$ generated by a basis (v_1, v_2) of $\mathbb{R}^2$ is the subgroup $\Lambda = \{jv_1 + kv_2 : (j, k) \in \mathbb{Z}^2\}$. Its **covolume** $|\Lambda| = |\det(v_1, v_2)|$ is the area of its fundamental parallelogram, and its asymptotic density is $1/|\Lambda|$.

Definition 1.4 (The density-one triangular lattice A)**.** Let $b = \frac{\sqrt{3}}{2}$. Define the basis vectors

$$v_1 = \frac{1}{\sqrt{b}}(1,\, 0)\,, \qquad v_2 = \frac{1}{\sqrt{b}}\left(\frac{1}{2},\, b\right),$$

and the lattice map $p : \mathbb{Z}^2 \to \mathbb{R}^2$ by

$$p(j, k) = jv_1 + kv_2 = \frac{1}{\sqrt{b}}\left(j + \frac{k}{2},\, kb\right).$$

The **covolume-one triangular lattice** is the set $A = \{p(j, k) : (j, k) \in \mathbb{Z}^2\}$.

Proposition 1.5 (Covolume, Gram matrix, and squared-norm identity)**.** *The lattice A has covolume $|A| = 1$ (hence density one). Its Gram matrix in the basis (v_1, v_2) is*

$$G = \begin{pmatrix} \langle v_1, v_1 \rangle & \langle v_1, v_2 \rangle \\ \langle v_2, v_1 \rangle & \langle v_2, v_2 \rangle \end{pmatrix} = \frac{1}{b} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix},$$

and for every $(j, k) \in \mathbb{Z}^2$, the squared Euclidean norm of $p(j, k)$ is

$$\|p(j,k)\|^2 = \frac{1}{b}\left(j^2 + jk + k^2\right) = \frac{2}{\sqrt{3}}\left(j^2 + jk + k^2\right). \tag{1}$$

Proof. First, the determinant of the basis (v_1, v_2) in the canonical basis of $\mathbb{R}^2$ is

$$\det(v_1, v_2) = \left(\frac{1}{\sqrt{b}}\right)^2 \det \begin{pmatrix} 1 & 1/2 \\ 0 & b \end{pmatrix} = \frac{1}{b} \cdot b = 1.$$

Next, we compute the scalar products: $\langle v_1, v_1 \rangle = \frac{1}{b}$, $\langle v_1, v_2 \rangle = \frac{1}{b}\left(1 \cdot \frac{1}{2} + 0\right) = \frac{1}{2b}$, and since $b^2 = \frac{3}{4}$,

$$\langle v_2, v_2 \rangle = \frac{1}{b}\left(\frac{1}{4} + b^2\right) = \frac{1}{b}\left(\frac{1}{4} + \frac{3}{4}\right) = \frac{1}{b}.$$

Thus, expanding $\|jv_1 + kv_2\|^2$ gives

$$\|p(j,k)\|^2 = \begin{pmatrix} j & k \end{pmatrix} G \begin{pmatrix} j \\ k \end{pmatrix} = \frac{1}{b}\left(j^2 + jk + k^2\right) = \frac{2}{\sqrt{3}}\left(j^2 + jk + k^2\right). \qquad \square$$

Definition 1.6 (Dual lattice)**.** The **dual lattice** of a lattice $\Lambda \subset \mathbb{R}^2$ is

$$\Lambda^* = \{w \in \mathbb{R}^2 : \forall \lambda \in \Lambda,\ \langle w, \lambda \rangle \in \mathbb{Z}\}.$$

Proposition 1.7 (A^* is a $90°$ rotation of A)**.** *Let $J : \mathbb{R}^2 \to \mathbb{R}^2$ be the counter-clockwise rotation by $90°$, namely $J(x_1, x_2) = (-x_2, x_1)$. Then $A^* = J(A)$. In particular, A^* is isometric to A and has covolume $|A^*| = 1$.*

Proof. Apply J to the basis vectors of A:

$$Jv_1 = \frac{1}{\sqrt{b}}(0, 1), \qquad Jv_2 = \frac{1}{\sqrt{b}}\left(-b, \frac{1}{2}\right).$$

Computing the four scalar products with v_1 and v_2 yields:

$$\langle Jv_1, v_1 \rangle = 0, \quad \langle Jv_1, v_2 \rangle = \frac{b}{b} = 1, \quad \langle Jv_2, v_1 \rangle = -\frac{b}{b} = -1, \quad \langle Jv_2, v_2 \rangle = \frac{1}{b}\left(-\frac{b}{2} + \frac{b}{2}\right) = 0.$$

Since all four scalar products are in $\mathbb{Z}$, bilinearity shows that every integer combination of Jv_1 and Jv_2 has integer inner product with every point of A, so $J(A) \subseteq A^*$. Moreover, these relations say exactly that $Jv_1 = v_2^*$ and $Jv_2 = -v_1^*$, where (v_1^*, v_2^*) is the dual basis of (v_1, v_2) (Appendix A); so (Jv_1, Jv_2) is the dual basis up to sign and order, and since J is a rotation, $|\det(Jv_1, Jv_2)| = |\det(v_1, v_2)| = 1 = |A^*|$. Because $J(A)$ is a sublattice of A^* of the exact same covolume, $J(A) = A^*$. $\square$

1.3 Admissible Potentials and Lower Energy

Following Cohn–Kumar [CK07] and CKMRV [CKMRV22], we write any pair potential as a function $g(t)$ of the **squared distance** $t = \|x-y\|^2$. Thus, an interaction $V(r)$ at distance $r = \|x-y\|$ corresponds to $g(t) = V(\sqrt{t})$.

Definition 1.8 (Completely monotone / admissible potential)**.** A function $g : (0,\infty) \to \mathbb{R}$ is **completely monotone** (called **admissible** in the formalization) if:

(i) g is smooth of class C^∞ on $(0,\infty)$;

(ii) $g(t) \geq 0$ for all $t > 0$;

(iii) For every integer $r \geq 0$ and every $t > 0$,

$$(-1)^r g^{(r)}(t) \geq 0.$$

(Condition (ii) is the case $r = 0$ of (iii); it is kept for readability.)

Definition 1.9 (Disk energy, lower energy, and lattice energy)**.** Let $g : (0,\infty) \to [0,\infty)$ and let $C \subseteq \mathbb{R}^2$ be locally finite. We work in the extended non-negative real numbers $[0,\infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$ with the conventions $1/0 = \infty$ and $\infty \cdot 0 = 0$.

(i) The **disk energy** of C at radius R is the ordered-pair average

$$E_g(C;R) = \frac{1}{N_R(C)} \sum_{x \in C_R} \sum_{y \in C_R \setminus \{x\}} g\left(\|x-y\|^2\right) \in [0,\infty]. \tag{2}$$

(ii) The **lower energy** of C is the limit inferior over all radii $R \to \infty$:

$$E_g(C) = \liminf_{R\to\infty} E_g(C;R) = \sup_{R_0>0} \inf_{R \geq R_0} E_g(C;R) \in [0,\infty]. \tag{3}$$

(iii) The **lattice energy** of the triangular lattice A is the series seen from the origin:

$$\mathcal{E}_g(A) = \sum_{a \in A\setminus\{0\}} g\left(\|a\|^2\right) = \sum_{(j,k)\in\mathbb{Z}^2\setminus\{(0,0)\}} g\left(\frac{2}{\sqrt{3}}(j^2+jk+k^2)\right) \in [0,\infty]. \tag{4}$$

Theorem 1.10 (Main Theorem: 2D Universal Energy Minimum — conditional on Imported Claim 2.6)**.** *Assume Imported Claim 2.6. Let g be an admissible potential (Definition 1.8) and let $C \subseteq \mathbb{R}^2$ be any locally finite configuration of centred density one. Then*

$$\mathcal{E}_g(A) \leq E_g(C) \qquad \textit{and} \qquad \mathcal{E}_g(A) = E_g(A) \qquad \textit{in } [0,\infty].$$

Sections 2 and 3 give the human mathematical derivation of Theorem 1.10 from Imported Claim 2.6; Section 2.4 states precisely what that claim asserts and why it is not proved here.

2 The Linear Programming Bound and the 2D Auxiliary Function

We normalize the Fourier transform of a Schwartz function $f \in \mathcal{S}(\mathbb{R}^2)$ as

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i \langle x,\xi\rangle}\, dx.$$

For $\alpha > 0$, we denote the Gaussian pair potential by $g_\alpha(t) = e^{-\pi\alpha t}$, so that $G_\alpha(x) := g_\alpha(\|x\|^2) = e^{-\pi\alpha\|x\|^2}$. Its Fourier transform is $\widehat{G_\alpha}(\xi) = \frac{1}{\alpha} e^{-\pi\|\xi\|^2/\alpha} > 0$ (proved in Appendix E).

2.1 The Sharp Auxiliary ("Magic") Function Conditions

Definition 2.1 (Sharp auxiliary function for G_α)**.** Let $\alpha > 0$. A radial Schwartz function $f_\alpha : \mathbb{R}^2 \to \mathbb{R}$ is an **auxiliary function** for G_α if:

(i) **Spatial minorant:** $f_\alpha(x) \le G_\alpha(x)$ for all $x \in \mathbb{R}^2 \setminus \{0\}$;

(ii) **Fourier non-negativity:** $\widehat{f_\alpha}(\xi) \ge 0$ for all $\xi \in \mathbb{R}^2$.

We say that f_α is **sharp for the triangular lattice** A if in addition it satisfies the contact conditions:

(iii) **Primal contact on** A**:** $f_\alpha(a) = G_\alpha(a)$ for all $a \in A \setminus \{0\}$;

(iv) **Dual roots on** A^***:** $\widehat{f_\alpha}(w) = 0$ for all $w \in A^* \setminus \{0\}$.

2.2 Complete Proof of the LP Bound for Periodic Configurations

To see clearly how Definition 2.1 forces A to be optimal, we first prove the bound in full detail for arbitrary periodic configurations.

Definition 2.2 (Periodic configuration and periodic energy)**.** Let $\Lambda \subset \mathbb{R}^2$ be a lattice and let $v_1, \dots, v_N \in \mathbb{R}^2$ represent distinct cosets ($v_m - v_n \notin \Lambda$ for $m \ne n$). The configuration $C = \bigcup_{m=1}^N (\Lambda + v_m)$ is Λ-periodic with N points per fundamental domain, and has density $\rho = N/|\Lambda|$. Its **periodic energy** per particle is

$$E_g^{\mathrm{per}}(C) = \frac{1}{N} \sum_{m=1}^{N} \sum_{n=1}^{N} \sum_{\substack{\lambda \in \Lambda \\ \lambda + v_n - v_m \ne 0}} g\left(\|\lambda + v_n - v_m\|^2\right).$$

Theorem 2.3 (Cohn–Kumar LP Bound [CK07], Periodic Case)**.** *Let $\alpha > 0$, let f_α be an auxiliary function for G_α satisfying (i)–(ii) of Definition 2.1, and let $C = \bigcup_{m=1}^N (\Lambda + v_m)$ be any periodic configuration of density one (so $|\Lambda| = N$). Then*

$$E_{g_\alpha}^{per}(C) \ge \widehat{f_\alpha}(0) - f_\alpha(0). \tag{5}$$

If furthermore f_α is sharp for A (satisfying (iii)–(iv)), then equality holds for $C = A$:

$$E_{g_\alpha}^{per}(A) = \widehat{f_\alpha}(0) - f_\alpha(0) = \sum_{a \in A \setminus \{0\}} e^{-\pi\alpha\|a\|^2} = \mathcal{E}_{g_\alpha}(A).$$

Proof. Because G_α and f_α are in $\mathcal{S}(\mathbb{R}^2)$, all lattice sums below converge absolutely.
Step 1 (Minorant inequality): Since $\lambda + v_n - v_m \ne 0$ for every term in the sum, Condition (i) ($f_\alpha(x) \le G_\alpha(x)$ for $x \ne 0$) applied term-by-term gives

$$N E_{g_\alpha}^{\mathrm{per}}(C) = \sum_{m,n=1}^{N} \sum_{\lambda + v_n - v_m \ne 0} G_\alpha(\lambda + v_n - v_m) \ge \sum_{m,n=1}^{N} \sum_{\lambda + v_n - v_m \ne 0} f_\alpha(\lambda + v_n - v_m).$$

Step 2 (Restoring the diagonal): Since $v_m - v_n \notin \Lambda$ for $m \ne n$, the vector $\lambda + v_n - v_m$ vanishes if and only if $m = n$ and $\lambda = 0$. There are exactly N such excluded diagonal terms, each equal to $f_\alpha(0)$. Adding and subtracting them yields

$$\sum_{m,n=1}^{N} \sum_{\lambda + v_n - v_m \ne 0} f_\alpha(\lambda + v_n - v_m) = \left(\sum_{m,n=1}^{N} \sum_{\lambda \in \Lambda} f_\alpha(\lambda + v_n - v_m) \right) - N f_\alpha(0).$$

Step 3 (2D Poisson Summation): By the Poisson Summation Formula on Λ (Theorem E.2 in Appendix E) applied to the shift $v = v_n - v_m$ with covolume $|\Lambda| = N$,

$$\sum_{\lambda\in\Lambda} f_\alpha(\lambda + v_n - v_m) = \frac{1}{N}\sum_{w\in\Lambda^*} \widehat{f_\alpha}(w) e^{2\pi i\langle w, v_n - v_m\rangle}.$$

Summing over the finite indices $m, n \in \{1, \dots, N\}$ and factoring the exponential sum gives

$$\sum_{m,n=1}^{N}\sum_{\lambda\in\Lambda} f_\alpha(\lambda+v_n-v_m) = \frac{1}{N}\sum_{w\in\Lambda^*}\widehat{f_\alpha}(w)\sum_{m,n=1}^{N} e^{2\pi i\langle w,v_n\rangle}e^{-2\pi i\langle w,v_m\rangle} = \frac{1}{N}\sum_{w\in\Lambda^*}\widehat{f_\alpha}(w)\left|\sum_{n=1}^{N} e^{2\pi i\langle w,v_n\rangle}\right|^2.$$

Step 4 (Fourier positivity): By Condition (ii), $\widehat{f_\alpha}(w) \geq 0$ for every $w \in \Lambda^*$, and the squared modulus is non-negative. Dropping all terms with $w \neq 0$ lowers the sum to the $w = 0$ term alone, where $\sum_{n=1}^{N} e^0 = N$:

$$\frac{1}{N}\sum_{w\in\Lambda^*}\widehat{f_\alpha}(w)\left|\sum_{n=1}^{N} e^{2\pi i\langle w,v_n\rangle}\right|^2 \geq \frac{1}{N}\widehat{f_\alpha}(0)N^2 = N\widehat{f_\alpha}(0).$$

Combining Steps 1–4 and dividing by N proves $E^{\mathsf{per}}_{g_\alpha}(C) \geq \widehat{f_\alpha}(0) - f_\alpha(0)$.
Step 5 (Sharpness and attainment on A): Take $\Lambda = A$, $N = 1$, and $v_1 = 0$. In Step 1, Condition (iii) ($f_\alpha(a) = G_\alpha(a)$ on $A\setminus\{0\}$) turns the minorant inequality into an exact equality. In Step 4, Condition (iv) ($\widehat{f_\alpha}(w) = 0$ on $A^*\setminus\{0\}$) makes all dropped $w \neq 0$ terms identically zero. Thus both inequalities become equalities:

$$\mathcal{E}_{g_\alpha}(A) = \sum_{a\in A\setminus\{0\}} G_\alpha(a) = \widehat{f_\alpha}(0) - f_\alpha(0). \qquad \square$$

2.3 Connecting Periodic Energy to Lower Disk Energy

Proposition 2.4 (Disk energy of a periodic configuration)**.** *For any Λ-periodic configuration $C = \bigcup_{m=1}^{N}(\Lambda + v_m)$ and any $\alpha > 0$,*

$$\lim_{R\to\infty} E_{g_\alpha}(C;R) = E^{\mathsf{per}}_{g_\alpha}(C).$$

In particular, the lower energy $E_{g_\alpha}(C)$ equals $E^{\mathsf{per}}_{g_\alpha}(C)$ and the limit exists.

Proof. For each $x \in C$, let $e(x) = \sum_{y\in C\setminus\{x\}} G_\alpha(x-y)$. By translation invariance of Λ, $e(x)$ is constant on each coset $\Lambda + v_m$, taking a value e_m, and $E^{\mathsf{per}}_{g_\alpha}(C) = \frac{1}{N}\sum_{m=1}^{N} e_m$. Let $n_m(R) = \mathrm{card}((\Lambda+v_m)\cap\overline{B}(0,R))$. By standard lattice point counting (Lemma A.1), $n_m(R)/(\pi R^2) \to 1/|\Lambda|$, so $n_m(R)/N_R(C) \to 1/N$ as $R\to\infty$.
Upper bound: Since $G_\alpha > 0$, replacing the inner sum over $y \in C_R\setminus\{x\}$ by all $y \in C\setminus\{x\}$ yields

$$E_{g_\alpha}(C;R) \leq \frac{1}{N_R(C)}\sum_{x\in C_R} e(x) = \sum_{m=1}^{N}\frac{n_m(R)}{N_R(C)}e_m \xrightarrow[R\to\infty]{} \frac{1}{N}\sum_{m=1}^{N} e_m = E^{\mathsf{per}}_{g_\alpha}(C).$$

Lower bound: For any $L > 0$, define the Gaussian tail $\tau(L) = \sup_{u\in\mathbb{R}^2}\sum_{\lambda\in\Lambda,\,\|\lambda+u\|>L} G_\alpha(\lambda+u)$. By rapid Gaussian decay, $\tau(L) < \infty$ and $\lim_{L\to\infty}\tau(L) = 0$. Fix $L > 0$ and take $R > L$. For any $x \in C_{R-L}$, the triangle inequality implies that every $y \in C$ with $\|x-y\| \leq L$ satisfies $\|y\| \leq R$, so $y \in C_R$. Thus the inner sum over $y \in C_R\setminus\{x\}$ misses at most points at distance $> L$ in each of the N cosets, so it is at least $e(x) - N\tau(L)$. Restricting the outer sum to $x \in C_{R-L}$ (the dropped inner sums are non-negative) gives

$$E_{g_\alpha}(C;R) \geq \frac{1}{N_R(C)}\sum_{x\in C_{R-L}}(e(x) - N\tau(L)) = \sum_{m=1}^{N}\frac{n_m(R-L)}{N_R(C)}\left(e_m - N\tau(L)\right).$$

Since $\frac{(R-L)^2}{R^2} \to 1$, we have $\frac{n_m(R-L)}{N_R(C)} \to \frac{1}{N}$ as $R \to \infty$. Taking $\liminf_{R\to\infty}$ yields

$$\liminf_{R\to\infty} E_{g_\alpha}(C;R) \geq E^{\mathrm{per}}_{g_\alpha}(C) - N\tau(L).$$

Letting $L \to \infty$ gives $\liminf_{R\to\infty} E_{g_\alpha}(C;R) \geq E^{\mathrm{per}}_{g_\alpha}(C)$, completing the proof. □

2.4 What Remains: Non-Periodic Competitors and the 2D Auxiliary Function (Not Proved Here)

Theorem 2.3 yields the bound only for *periodic* competitors, and only once an auxiliary function f_α is in hand. The upstream theorem requires two further ingredients. The first is a standard result of the literature that we cite rather than reproduce; the second is the whole substance of the upstream claim and is *not* verified in this document.

1. **Step A — LP bound for arbitrary locally finite configurations (standard, cited).**

   Proposition 2.5 (LP bound, general case [CdCI18, Prop. 2.2])**.** *Let $\alpha > 0$ and let f_α satisfy (i)–(ii) of Definition 2.1. Then for every locally finite configuration $C \subseteq \mathbb{R}^2$ of centred density one,*

   $$E_{g_\alpha}(C) \geq \widehat{f_\alpha}(0) - f_\alpha(0).$$

   This is exactly the Cohn–Kumar bound of Theorem 2.3 extended to non-periodic configurations with the lower energy of Definition 1.9; Cohn and de Courcy-Ireland prove it under even weaker hypotheses (continuity, integrability, and positive-definiteness of the auxiliary function suffice). The mechanism, for the reader who wants to audit the source: Fourier positivity applied to the point masses alone gives only

   $$\sum_{x,y\in C_R} f_\alpha(x-y) = \int_{\mathbb{R}^2} \widehat{f_\alpha}(\xi) \left| \sum_{x\in C_R} e^{-2\pi i\langle x,\xi\rangle} \right|^2 d\xi \geq 0,$$

   i.e. the useless bound $E_{g_\alpha}(C;R) \geq -f_\alpha(0)$. The missing term $\widehat{f_\alpha}(0)$ is produced by applying positive-definiteness instead to the *signed* measure

   $$\nu_R = \sum_{x\in C_R} \delta_x - \mathbf{1}_{\overline{B}(0,R)}(x)\,dx$$

   (the points minus a uniform background of density one). Expanding the non-negative quantity $\iint f_\alpha(x-y)\,d\nu_R(x)\,d\nu_R(y)$ gives

   $$\sum_{x\neq y\in C_R} f_\alpha(x-y) \;\geq\; -N_R f_\alpha(0) \;+\; 2\sum_{x\in C_R} \int_{\overline{B}(0,R)} f_\alpha(x-y)\,dy \;-\; \iint_{\overline{B}(0,R)^2} f_\alpha(x-y)\,dx\,dy,$$

   where the two integrals equal $N_R\widehat{f_\alpha}(0)+o(R^2)$ and $\pi R^2\widehat{f_\alpha}(0)+o(R^2)$ respectively (the errors come from the boundary annulus), and the density hypothesis $N_R/(\pi R^2) \to 1$ together with the $\liminf$ absorbs the error terms. The argument is elementary but delicate and is not reproduced here.

2. **Step B — Existence of sharp 2D auxiliary functions (imported, unverified).**

   Imported Claim 2.6 (Sharp 2D auxiliary functions for all Gaussians, [OAI26])**.** *For every $\alpha > 0$ there exists a radial Schwartz function $f_\alpha : \mathbb{R}^2 \to \mathbb{R}$ satisfying all four conditions (i)–(iv) of Definition 2.1. Consequently (by Proposition 2.5 and Step 5 of Theorem 2.3), for every $\alpha > 0$ and every locally finite configuration $C \subseteq \mathbb{R}^2$ of centred density one,*

   $$E_{g_\alpha}(C) \geq \widehat{f_\alpha}(0) - f_\alpha(0) = \mathcal{E}_{g_\alpha}(A) = \sum_{a\in A\setminus\{0\}} e^{-\pi\alpha\|a\|^2}. \tag{6}$$

This is the entire difficulty of the problem. The description below of how the upstream release constructs f_α is a *description of the upstream argument*, not a verification of it; the construction is AI-generated and has not been peer reviewed [XC26].

- *Why the 8D and 24D method does not transfer directly to 2D (heuristic):* In dimensions 8 (E_8) and 24 (the Leech lattice Λ_{24}), the squared norms of nonzero lattice vectors are exactly the even integers $2n$ with $n \geq 1$ (for E_8) and $n \geq 2$ (for Λ_{24}, which has no vectors of norm 2). This uniform spacing is what allows the auxiliary functions to be built from modular and quasimodular forms [Via17, CKMRV22]. In 2D, by Proposition 1.5, the squared norms of A are $\frac{2}{\sqrt{3}}n$ where $n = j^2 + jk + k^2$ runs over the Löschian numbers $\{1, 3, 4, 7, 9, 12, 13, \dots\}$. Because integers such as $n = 2, 5, 6, 8, 10, 11$ do not occur, the lattice shells have arithmetic gaps, and no modular-form construction of a sharp f_α is known. (This is a heuristic explanation of the obstruction, not a theorem.)
- *Double-zero contact structure:* Since $G_\alpha(x) - f_\alpha(x) \geq 0$ everywhere on $\mathbb{R}^2 \setminus \{0\}$ and vanishes at every lattice shell $a \in A \setminus \{0\}$, every point $a \in A \setminus \{0\}$ is a local minimum of $G_\alpha - f_\alpha$. Thus its gradient vanishes ($\nabla f_\alpha(a) = \nabla G_\alpha(a)$), and as a function of squared radius $t = \|x\|^2$, $g_\alpha(t) - \tilde{f}_\alpha(t)$ has **double zeros** at all lattice shells (Appendix F).
- *Sine-product cardinal interpolation and ball arithmetic (upstream's construction, as described):* Instead of modular forms, upstream constructs f_α from a **sine product with double zeros**, multiplied by simple-pole and double-pole rational factors to prescribe exact values and slopes on the initial shells of A and A^*. Applying Gaussian damping and compactly supported spectral measures reduces the verification of $f_\alpha \leq G_\alpha$ and $\widehat{f_\alpha} \geq 0$ to finite reference lists handled by positive quadrature and certified via rigorous interval/ball arithmetic (`python-flint`). *Audit note for reviewers:* in the variable $z = \frac{\sqrt{3}}{2}t$, a squared sine product $\sin^2(\pi z)$ has double zeros at *every* integer $z = n$, not only at the Löschian ones; the upstream construction therefore forces contact $f_\alpha = G_\alpha$ at non-lattice shells ($n = 2, 5, 6, \dots$) as well. Extra contact points are permitted by Definition 2.1, but a reviewer should check how the upstream manuscript reconciles them with the dual roots on $A^* = J(A)$ (which has the *same* set of shells) and with $\widehat{f_\alpha} \geq 0$.

3 From Gaussians to All Completely Monotone Potentials

Assume Imported Claim 2.6, i.e. the Gaussian lower bound (6): for every $\alpha > 0$ and every locally finite configuration C of density one,

$$E_{g_\alpha}(C) \geq \mathcal{E}_{g_\alpha}(A) = \sum_{a \in A \setminus \{0\}} e^{-\pi\alpha\|a\|^2}.$$

We now prove that the inequality $\mathcal{E}_g(A) \leq E_g(C)$ and attainment $\mathcal{E}_g(A) = E_g(A)$ hold for **every** admissible (completely monotone) potential g. Everything in this section is proved in full.

3.1 The Constant Potential Lemma (`energy_one`)

Before integrating over Gaussian mixtures, we must handle the constant potential $g \equiv 1$ (which corresponds to the exponent $\alpha = 0$).

Lemma 3.1 (Infinite energy of a constant potential)**.** *Let $C \subseteq \mathbb{R}^2$ be locally finite with density one. For the constant potential $g \equiv 1$,*

$$E_1(C) = \infty \qquad \textit{and} \qquad \mathcal{E}_1(A) = \infty.$$

Proof. For any radius R with $N_R(C) \geq 1$, the disk energy of $g \equiv 1$ is

$$E_1(C;R) = \frac{1}{N_R(C)} \sum_{x \in C_R} \sum_{y \in C_R \setminus \{x\}} 1 = \frac{N_R(C)\,(N_R(C)-1)}{N_R(C)} = N_R(C) - 1.$$

Since C has density one, $N_R(C) \sim \pi R^2 \to \infty$ as $R \to \infty$. Therefore $E_1(C) = \liminf_{R\to\infty}(N_R(C) - 1) = \infty$. Likewise, $\mathcal{E}_1(A) = \sum_{a \in A\setminus\{0\}} 1 = \infty$ since $A \setminus \{0\}$ is infinite. □

3.2 Proof of Theorem 1.10 via Bernstein's Theorem, Fatou, and Tonelli

Theorem 3.2 (Bernstein–Hausdorff–Widder Representation [Ber29], [Wid41, Ch. IV])**.** *Every completely monotone function $g : (0,\infty) \to [0,\infty)$ can be uniquely represented as the Laplace transform of a non-negative Borel measure μ on $[0,\infty)$:*

$$g(t) = \int_{[0,\infty)} e^{-\pi\alpha t}\, d\mu(\alpha) \qquad \forall t > 0. \tag{7}$$

We do not reprove Theorem 3.2; note that μ may have infinite total mass (e.g. $g(t) = t^{-1}$ corresponds to $d\mu(\alpha) = \pi\, d\alpha$), but the Laplace integral converges for every $t > 0$.

Complete Proof of Theorem 1.10 from the Gaussian Bounds. Let g be admissible and let C be locally finite of density one. By Theorem 3.2, write $g(t) = \int_{[0,\infty)} g_\alpha(t)\, d\mu(\alpha)$ where $g_\alpha(t) = e^{-\pi\alpha t}$ and μ is a non-negative Borel measure on $[0,\infty)$.
Step 1 (Commuting finite disk sums with the integral): Fix any radius $R > 0$ with $N_R(C) \geq 1$. Because the double sum over $x \neq y \in C_R$ is finite, linearity of the integral gives

$$E_g(C;R) = \frac{1}{N_R(C)} \sum_{x \neq y \in C_R} \int_{[0,\infty)} g_\alpha\left(\|x-y\|^2\right) d\mu(\alpha) = \int_{[0,\infty)} E_{g_\alpha}(C;R)\, d\mu(\alpha).$$

Step 2 (Passing to the lim inf **via Fatou's Lemma):** The limit inferior is *not* linear, so we cannot pull $\liminf_{R\to\infty}$ inside the integral by linearity. Instead, choose a sequence of radii $R_n \to \infty$ that realizes the limit inferior of the total energy:

$$\lim_{n\to\infty} E_g(C;R_n) = \liminf_{R\to\infty} E_g(C;R) = E_g(C).$$

Since $E_{g_\alpha}(C;R_n) \geq 0$ for all $\alpha \geq 0$ and all $n \in \mathbb{N}$ (and $\alpha \mapsto E_{g_\alpha}(C;R_n)$ is continuous, hence measurable), we apply **Fatou's Lemma** (Theorem C.1 in Appendix C) to the sequence of non-negative measurable functions $\alpha \mapsto E_{g_\alpha}(C;R_n)$:

$$E_g(C) = \liminf_{n\to\infty} \int_{[0,\infty)} E_{g_\alpha}(C;R_n)\, d\mu(\alpha) \geq \int_{[0,\infty)} \left(\liminf_{n\to\infty} E_{g_\alpha}(C;R_n)\right) d\mu(\alpha).$$

Because the lim inf along a subsequence (R_n) is always greater than or equal to the lim inf over all real radii $R \to \infty$, we have $\liminf_{n\to\infty} E_{g_\alpha}(C;R_n) \geq E_{g_\alpha}(C)$ for every $\alpha \geq 0$. Hence

$$E_g(C) \geq \int_{[0,\infty)} E_{g_\alpha}(C)\, d\mu(\alpha). \tag{8}$$

Step 3 (Handling the atom at $\alpha = 0$): Split $[0,\infty) = \{0\} \cup (0,\infty)$.

- *Case A ($\mu(\{0\}) = c > 0$):* Since $g_\alpha(t) \geq 0$, we have $g(t) \geq c \cdot g_0(t) = c$ for all $t > 0$. By Lemma 3.1, $E_g(C) \geq cE_1(C) = c \cdot \infty = \infty$, and $\mathcal{E}_g(A) = \infty$, so $\mathcal{E}_g(A) \leq E_g(C)$ holds as $\infty \leq \infty$.

- *Case B ($\mu(\{0\}) = 0$):* Applying the Gaussian optimality bound (6) for every $\alpha \in (0, \infty)$ inside (8) gives

$$E_g(C) \geq \int_{(0,\infty)} E_{g_\alpha}(C)\, d\mu(\alpha) \geq \int_{(0,\infty)} \left(\sum_{a \in A \setminus \{0\}} e^{-\pi\alpha\|a\|^2} \right) d\mu(\alpha).$$

Step 4 (Swapping sum and integral via Tonelli's Theorem): Since every term $e^{-\pi\alpha\|a\|^2}$ is non-negative and measurable on the product space $(A \setminus \{0\}) \times (0, \infty)$ (where $A \setminus \{0\}$ carries the counting measure), **Tonelli's Theorem** (Theorem C.2 in Appendix C) justifies interchanging the series and the integral without any a priori finiteness assumption:

$$\int_{(0,\infty)} \left(\sum_{a \in A \setminus \{0\}} e^{-\pi\alpha\|a\|^2} \right) d\mu(\alpha) = \sum_{a \in A \setminus \{0\}} \int_{(0,\infty)} e^{-\pi\alpha\|a\|^2}\, d\mu(\alpha) = \sum_{a \in A \setminus \{0\}} g\left(\|a\|^2\right) = \mathcal{E}_g(A),$$

where the last equality uses $\mu(\{0\}) = 0$ (Case B). Thus $E_g(C) \geq \mathcal{E}_g(A)$.
Step 5 (Attainment on A): Since A is locally finite of density one, applying the bound just proved to $C = A$ gives $\mathcal{E}_g(A) \leq E_g(A) = \liminf_{R\to\infty} E_g(A; R)$. Conversely, for every $R > 0$ and every $x \in A_R$, the set $\{y - x : y \in A_R \setminus \{x\}\}$ is a subset of $A \setminus \{0\}$ (because A is a subgroup under subtraction). Since $g \geq 0$,

$$\sum_{y \in A_R \setminus \{x\}} g\left(\|x - y\|^2\right) \leq \sum_{a \in A \setminus \{0\}} g\left(\|a\|^2\right) = \mathcal{E}_g(A).$$

Averaging over the $N_R(A)$ points $x \in A_R$ gives $E_g(A; R) \leq \mathcal{E}_g(A)$ for every $R > 0$. Taking the lim inf as $R \to \infty$ gives $E_g(A) \leq \mathcal{E}_g(A)$. Combining the two inequalities yields $E_g(A) = \mathcal{E}_g(A)$ (in Case A, both sides are ∞). □

Remark 3.3 (Upstream's shifted-mixture formalization route)*.* In the Lean 4 formalization, to avoid invoking Bernstein's theorem on the open half-line $(0, \infty)$ where $g(t)$ may blow up as $t \to 0^+$, upstream considers for each $\epsilon > 0$ the shifted potential $g_\epsilon(t) = g(\epsilon + t)$, which is bounded on $[0, \infty)$ by $g(\epsilon)$. A finite-difference construction represents $g(\epsilon + t) = \int_{[0,1]} v^t\, d\nu_\epsilon(v)$ for a finite measure ν_ϵ on $[0, 1]$. Here $v \in (0, 1)$ corresponds to the Gaussian $v^t = e^{-\pi\alpha t}$ with $\alpha = -(\ln v)/\pi > 0$, $v = 1$ is the constant potential $1^t = 1$ (handled by Lemma 3.1), and $v = 0$ is the zero potential. Since g is non-increasing ($g' \leq 0$), $E_g(C) \geq E_{g_\epsilon}(C) \geq \mathcal{E}_{g_\epsilon}(A)$, and taking the supremum over $\epsilon = \frac{1}{n+1} \to 0$ recovers $\mathcal{E}_g(A)$ by monotone convergence.

4 Complete Proofs of the Four Corollaries (Results D1–D4)

Every result in this section is deduced from Theorem 1.10; it therefore inherits the conditional status of that theorem (see the epistemic-status box on page 1), but the deductions themselves are complete.

4.1 Result D1: Riesz Potentials ($V(r) = r^{-s}$)

For $s > 0$, the Riesz pair interaction $V(r) = r^{-s}$ is written as a function of squared distance $t = r^2$ as

$$g_s(t) = t^{-s/2}.$$

Lemma 4.1 (Admissibility of Riesz potentials)**.** *For every $s > 0$, the potential $g_s(t) = t^{-s/2}$ is admissible (completely monotone on $(0, \infty)$). For $s < 0$, g_s is not admissible in general (in particular, $g_{-2}(t) = t$ is not admissible).*

Proof. Let $\sigma = s/2 > 0$. On $(0,\infty)$, $g_s(t) = \exp(-\sigma \ln t)$ is a composition of C^∞ functions, hence C^∞, and $g_s(t) > 0$. By induction on $r \in \mathbb{N}$, the r-th derivative of $t^{-\sigma}$ is

$$g_s^{(r)}(t) = (-\sigma)(-\sigma-1)\cdots(-\sigma-r+1)\, t^{-\sigma-r} = (-\sigma)_r^{\downarrow}\, t^{-\sigma-r},$$

where $(x)_r^{\downarrow} = \prod_{i=0}^{r-1}(x-i)$ is the descending Pochhammer symbol. Factoring -1 out of each of the r terms gives

$$(-\sigma)_r^{\downarrow} = (-1)^r \sigma(\sigma+1)\cdots(\sigma+r-1) = (-1)^r (\sigma)_r^{\uparrow},$$

where $(\sigma)_r^{\uparrow} = \prod_{i=0}^{r-1}(\sigma+i)$ is the ascending Pochhammer symbol. Multiplying by $(-1)^r$ cancels the sign:

$$(-1)^r g_s^{(r)}(t) = (-1)^{2r}(\sigma)_r^{\uparrow} t^{-\sigma-r} = \left(\prod_{i=0}^{r-1}(\sigma+i)\right) t^{-\sigma-r} > 0 \quad \forall t > 0.$$

Thus g_s is admissible. Conversely, for $s = -2$, $g_{-2}(t) = t$, so at $r = 1$ and $t = 1$ we have $(-1)^1 g'_{-2}(1) = -1 < 0$, violating Definition 1.8(iii). □

Theorem 4.2 (Result D1: Riesz optimality)**.** *For every $s > 0$ and every locally finite configuration $C \subseteq \mathbb{R}^2$ of density one,*

$$\sum_{a \in A\setminus\{0\}} \|a\|^{-s} \le E_{g_s}(C) \qquad \textit{in } [0,\infty].$$

Proof. Immediate by applying Theorem 1.10 to $g = g_s$, which is admissible by Lemma 4.1. □

4.2 Result D1b: Finiteness of the Riesz Lattice Energy for $s > 2$ and Divergence for $s \le 2$

Theorem 4.3 (Result D1b: Convergence threshold of the Epstein zeta sum)**.** *The Riesz lattice energy of the triangular lattice*

$$\mathcal{E}_{g_s}(A) = \sum_{a \in A\setminus\{0\}} \|a\|^{-s} = b^{s/2} \sum_{(j,k)\in\mathbb{Z}^2\setminus\{(0,0)\}} \left(j^2 + jk + k^2\right)^{-s/2}$$

*is **finite** ($\mathcal{E}_{g_s}(A) < \infty$) if and only if $s > 2$. For $0 < s \le 2$, $\mathcal{E}_{g_s}(A) = \infty$.*

Proof. Let $\sigma = s/2$ and let $Q(j,k) = j^2 + jk + k^2$.
Part 1 (Finiteness for $s > 2$, i.e., $\sigma > 1$): By the elementary inequality $|jk| \le \frac{1}{2}(j^2+k^2)$, we have

$$\frac{1}{2}\left(j^2+k^2\right) \le Q(j,k) \le \frac{3}{2}\left(j^2+k^2\right) \qquad \forall (j,k) \in \mathbb{Z}^2. \tag{9}$$

In particular, $Q(j,k) = 0 \iff (j,k) = (0,0)$, and for $(j,k) \ne (0,0)$,

$$Q(j,k)^{-\sigma} \le 2^{\sigma}\left(j^2+k^2\right)^{-\sigma}.$$

Partition $\mathbb{Z}^2 \setminus \{(0,0)\}$ into square shells $S_n = \{(j,k) \in \mathbb{Z}^2 : \max(|j|,|k|) = n\}$ for $n \ge 1$. The shell S_n contains exactly $(2n+1)^2 - (2n-1)^2 = 8n$ integer points, and every $(j,k) \in S_n$ satisfies $j^2 + k^2 \ge n^2$. Therefore,

$$\sum_{(j,k)\ne(0,0)} Q(j,k)^{-\sigma} \le 2^{\sigma} \sum_{n=1}^{\infty} \sum_{(j,k)\in S_n} \left(j^2+k^2\right)^{-\sigma} \le 2^{\sigma} \sum_{n=1}^{\infty} 8n \cdot \left(n^2\right)^{-\sigma} = 8 \cdot 2^{\sigma} \sum_{n=1}^{\infty} \frac{1}{n^{2\sigma-1}}.$$

Since $s > 2$, we have $2\sigma - 1 = s - 1 > 1$, so the Riemann p-series converges. Multiplying by $b^{\sigma} = (\sqrt{3}/2)^{s/2}$ proves $\mathcal{E}_{g_s}(A) < \infty$.

Part 2 (Divergence for $0 < s \le 2$): Using the upper bound in (9), every $(j,k) \in S_n$ satisfies $j^2 + k^2 \le 2n^2$, so $Q(j,k) \le 3n^2$. Thus for $0 < s \le 2$ ($\sigma > 0$),

$$\sum_{(j,k)\neq(0,0)} Q(j,k)^{-\sigma} \ge \sum_{n=1}^{\infty} \sum_{(j,k)\in S_n} \left(3n^2\right)^{-\sigma} = 8 \cdot 3^{-\sigma} \sum_{n=1}^{\infty} \frac{1}{n^{s-1}} = \infty$$

since $s - 1 \le 1$. (More precisely, by Stieltjes summation by parts against the disk count $N(r) = \pi r^2 + O(r)$, the partial sum over $0 < \|a\| \le R$ grows like $2\pi \ln R + O(1)$ for $s = 2$ and like $\frac{2\pi}{2-s} R^{2-s}$ for $0 < s < 2$.) □

4.3 Result D2: Universal Optimality at Every Density $\rho > 0$

We now transfer Theorem 1.10 from density one to any density $\rho > 0$ via four scaling lemmas. For $c > 0$ and $C \subseteq \mathbb{R}^2$, let $cC = \{cx : x \in C\}$.

Lemma 4.4 (Scaling of counts, density, energies, and potentials)**.** *Let $C \subseteq \mathbb{R}^2$ be locally finite, $c > 0$, $\rho > 0$, and $h : (0,\infty) \to \mathbb{R}$.*

(i) ***Counts:*** *cC is locally finite and $N_R(cC) = N_{R/c}(C)$ for all $R \in \mathbb{R}$.*

(ii) ***Density:*** *C has centred density ρ if and only if $\sqrt{\rho}C$ has centred density one.*

(iii) ***Energies:*** *For every $R \in \mathbb{R}$, $E_h(cC; R) = E_{h(c^2\cdot)}(C; R/c)$, and taking the* lim inf *yields*

$$E_h(cC) = E_{h(c^2\cdot)}(C).$$

(iv) ***Potentials:*** *If h is admissible, then the rescaled potential $h_c(t) = h(ct)$ is admissible.*

Proof. (i) For any $x \in \mathbb{R}^2$, $\|cx\| \le R \iff \|x\| \le R/c$. Thus $(cC) \cap \overline{B}(0,R) = c\left(C \cap \overline{B}(0,R/c)\right)$. Since $x \mapsto cx$ is a bijection, this set is finite of cardinality $N_{R/c}(C)$.
(ii) Taking $c = \sqrt{\rho}$ in (i) gives, for $R > 0$,

$$\frac{N_R(\sqrt{\rho}C)}{\pi R^2} = \frac{N_{R/\sqrt{\rho}}(C)}{\pi R^2} = \frac{1}{\rho} \cdot \frac{N_{R/\sqrt{\rho}}(C)}{\pi (R/\sqrt{\rho})^2}.$$

As $R \to \infty$, $R' = R/\sqrt{\rho} \to \infty$, so the right-hand side converges to $\frac{1}{\rho} \cdot \rho = 1$ if and only if $N_{R'}(C)/(\pi (R')^2) \to \rho$.
(iii) Under the bijection $x \mapsto cx$ between $C_{R/c}$ and $(cC)_R$, the counts agree and $\|cx - cy\|^2 = c^2 \|x - y\|^2$, so $E_h(cC; R) = E_{h(c^2\cdot)}(C; R/c)$. Since $R \mapsto R/c$ is an order-preserving bijection of $(0,\infty)$ onto itself, $\sup_{R_0} \inf_{R \ge R_0} F(R/c) = \sup_{R_0'} \inf_{R' \ge R_0'} F(R')$, so $E_h(cC) = E_{h(c^2\cdot)}(C)$.
(iv) If h is admissible and $c > 0$, $h_c(t) = h(ct)$ is C^∞ and non-negative on $(0,\infty)$. By induction and the chain rule, $h_c^{(r)}(t) = c^r h^{(r)}(ct)$, so $(-1)^r h_c^{(r)}(t) = c^r (-1)^r h^{(r)}(ct) \ge 0$. □

Theorem 4.5 (Result D2: Universal optimality at arbitrary density $\rho > 0$)**.** *Let $\rho > 0$, let h be an admissible potential, and let $C \subseteq \mathbb{R}^2$ be locally finite of centred density ρ. Then the rescaled triangular lattice $\rho^{-1/2} A$ is locally finite of density ρ, and*

$$E_h\left(\rho^{-1/2} A\right) = \sum_{a \in A \setminus \{0\}} h\left(\frac{\|a\|^2}{\rho}\right) \le E_h(C).$$

Proof. Let $c = \sqrt{\rho} > 0$ and define $h_{1/\rho}(t) = h(t/\rho)$, which is admissible by Lemma 4.4(iv). By Lemma 4.4(i)–(ii), $cC = \sqrt{\rho}C$ is locally finite of density one. Applying Theorem 1.10 to the potential $h_{1/\rho}$ and configuration $\sqrt{\rho}C$ gives

$$\mathcal{E}_{h_{1/\rho}}(A) = \sum_{a \in A \setminus \{0\}} h\left(\frac{\|a\|^2}{\rho}\right) \le E_{h_{1/\rho}}(\sqrt{\rho}C).$$

By Lemma 4.4(iii), $E_{h_{1/\rho}}(\sqrt{\rho}C) = E_{h_{1/\rho}(\rho\cdot)}(C) = E_h(C)$ since $h_{1/\rho}(\rho t) = h(\rho t/\rho) = h(t)$. Finally, $\rho^{-1/2}A$ has density ρ by Lemma 4.4(ii) (because $\sqrt{\rho} \cdot \rho^{-1/2}A = A$ has density one), and applying Lemma 4.4(iii) with $c = \rho^{-1/2}$ to A gives $E_h(\rho^{-1/2}A) = E_{h_{1/\rho}}(A) = \mathcal{E}_{h_{1/\rho}}(A)$ by the attainment part of Theorem 1.10. □

Remark 4.6 (Negative control: why the factor ρ^{-1} is essential)*.* For the Riesz potential $g_s(t) = t^{-s/2}$ ($s > 2$), homogeneity gives $\mathcal{E}_{g_s(\rho^{-1}\cdot)}(A) = \rho^{s/2}\mathcal{E}_{g_s}(A)$. At density $\rho = 1/4$ and $s = 4$, the rescaled lattice $2A = (1/4)^{-1/2}A$ has density $1/4$ and energy $E_{g_4}(2A) = (1/4)^2\mathcal{E}_{g_4}(A) = \frac{1}{16}\mathcal{E}_{g_4}(A) < \mathcal{E}_{g_4}(A)$ (since $0 < \mathcal{E}_{g_4}(A) < \infty$ by Theorem 4.3). Thus the unscaled density-one energy $\mathcal{E}_{g_4}(A)$ is strictly false as a lower bound at density $1/4$.

4.4 Result D3: The Vacancy Theorem and Non-Uniqueness of Minimizers

Theorem 1.10 identifies the *minimum energy value* $\mathcal{E}_g(A)$, not all configurations that achieve it. We now prove that removing the origin from the lattice yields a configuration that is universally optimal in the lower-energy sense, yet is not isometric to A.

Theorem 4.7 (Result D3: The Vacancy Theorem)**.** *Let $V = A \setminus \{0\}$ be the triangular lattice with one point removed (a single vacancy). Then:*

(i) V is locally finite and has centred density one;

(ii) For every admissible potential g, $E_g(V) = \mathcal{E}_g(A) = E_g(A)$ (so V is universally optimal at density one);

(iii) There exists no Euclidean isometry $f : \mathbb{R}^2 \to \mathbb{R}^2$ such that $f(A) = V$.

Proof. **(i) Local finiteness and density:** Since $V \subseteq A$ and A is locally finite, V is locally finite. For every $R \ge 0$, $0 \in A_R$, so $V_R = A_R \setminus \{0\}$ and $N_R(V) = N_R(A) - 1$. Therefore,

$$\frac{N_R(V)}{\pi R^2} = \frac{N_R(A)}{\pi R^2} - \frac{1}{\pi R^2} \xrightarrow[R\to\infty]{} 1 - 0 = 1.$$

(ii) Equality of lower energy: Let g be admissible. Since V is locally finite of density one by (i), Theorem 1.10 immediately gives the lower bound $\mathcal{E}_g(A) \le E_g(V)$. For the upper bound, let $S_V(R)$ and $S_A(R)$ denote the double sums over ordered pairs in V_R and A_R, respectively (both finite, since g is real-valued). Since $V_R \subset A_R$ and every term $g(\|x-y\|^2)$ is non-negative, $S_V(R) \le S_A(R)$. Because $N_R(A) \to \infty$, for all sufficiently large R we have $N_R(A) \ge 2$ and

$$E_g(V;R) = \frac{S_V(R)}{N_R(A)-1} \le \frac{N_R(A)}{N_R(A)-1}\cdot\frac{S_A(R)}{N_R(A)} = \kappa(R)E_g(A;R), \quad \text{where } \kappa(R) := \frac{N_R(A)}{N_R(A)-1} \xrightarrow[R\to\infty]{} 1.$$

Taking the limit inferior as $R \to \infty$ (using $\lim_{R\to\infty}\kappa(R) = 1 > 0$) yields

$$E_g(V) = \liminf_{R\to\infty} E_g(V;R) \le \liminf_{R\to\infty}(\kappa(R)E_g(A;R)) = 1 \cdot \liminf_{R\to\infty} E_g(A;R) = E_g(A) = \mathcal{E}_g(A).$$

Combining both inequalities proves $E_g(V) = \mathcal{E}_g(A)$.

(iii) Non-existence of an isometry: Suppose for contradiction that $f : \mathbb{R}^2 \to \mathbb{R}^2$ is an isometry satisfying $f(A) = V$. Every isometry of the Euclidean plane is an affine map ($f(x) = Ux + t$ for an orthogonal transformation $U \in O(2)$ and translation $t \in \mathbb{R}^2$). Consequently, f preserves affine midpoint reflections: for all $a, b \in \mathbb{R}^2$,

$$f(2a - b) = 2f(a) - f(b).$$

Consider the two nonzero lattice points $p(1, 0) = v_1$ and $p(2, 0) = 2v_1$. By Proposition 1.5, their squared norms are $1/b > 0$ and $4/b > 0$, so both belong to $V = A \setminus \{0\}$. Since $f(A) = V$, there exist $a, b \in A$ such that $f(a) = p(1, 0)$ and $f(b) = p(2, 0)$. Because A is a group under addition, $2a - b \in A$. However,

$$f(2a - b) = 2f(a) - f(b) = 2p(1, 0) - p(2, 0) = 2v_1 - 2v_1 = 0.$$

Thus $0 = f(2a - b) \in f(A) = V = A \setminus \{0\}$, which is a contradiction. □

4.5 Result D4: Equivalence of Open and Closed Disks

For a configuration $C \subseteq \mathbb{R}^2$, let $C_R^\circ = C \cap B^\circ(0, R)$ and $N_R^\circ(C) = \mathrm{card}(C_R^\circ)$ denote the point set and count in the **open** disk $B^\circ(0, R) = \{x : \|x\| < R\}$. Define the open-disk energy $E_g^\circ(C; R)$ and lower open-disk energy $E_g^\circ(C) = \liminf_{R\to\infty} E_g^\circ(C; R)$ identically to Definition 1.9 using C_R°.

Lemma 4.8 (Point-set comparison between open and closed disks)**.** *Let $C \subseteq \mathbb{R}^2$ be locally finite and let $R \in \mathbb{R}$. There exist radii $\varphi(R) \in [R - 1, R)$ and $\psi(R) \in (R, R + 1]$ such that*

$$C \cap B^\circ(0, R) = C \cap \overline{B}(0, \varphi(R)) \qquad \textit{and} \qquad C \cap \overline{B}(0, R) = C \cap B^\circ(0, \psi(R)).$$

Proof. First identity: The set $F = C \cap B^\circ(0, R)$ is a subset of $C \cap \overline{B}(0, R)$, so F is finite by local finiteness. Define $m = \max\{\|x\| : x \in F\}$ if $F \neq \emptyset$, and $m = -\infty$ if $F = \emptyset$. Since every $x \in F$ satisfies $\|x\| < R$ and F is finite, we have $m < R$. Set

$$\varphi(R) = \max\left(m,\, R - \frac{1}{2}\right) \in [R - 1, R).$$

If $x \in C \cap B^\circ(0, R) = F$, then $\|x\| \le m \le \varphi(R)$, so $x \in C \cap \overline{B}(0, \varphi(R))$. Conversely, if $x \in C \cap \overline{B}(0, \varphi(R))$, then $\|x\| \le \varphi(R) < R$, so $x \in C \cap B^\circ(0, R)$.
Second identity: The annulus set $G = \{x \in C : R < \|x\| \le R + 1\}$ is a subset of $C \cap \overline{B}(0, R + 1)$, so G is finite. Define $m' = \min\{\|x\| : x \in G\}$ if $G \neq \emptyset$, and $m' = +\infty$ if $G = \emptyset$. Since every $x \in G$ has $\|x\| > R$ and G is finite, $m' > R$. Set

$$\psi(R) = \min\left(m',\, R + \frac{1}{2}\right) \in (R, R + 1].$$

If $x \in C \cap \overline{B}(0, R)$, then $\|x\| \le R < \psi(R)$, so $x \in C \cap B^\circ(0, \psi(R))$. Conversely, suppose $x \in C \cap B^\circ(0, \psi(R))$. Then $\|x\| < \psi(R) \le R + \frac{1}{2} \le R + 1$. If we had $\|x\| > R$, then x would belong to G, forcing $\|x\| \ge m' \ge \psi(R)$, a contradiction. Thus $\|x\| \le R$, so $x \in C \cap \overline{B}(0, R)$. □

Lemma 4.9 (Lower limits under divergent reparametrization)**.** *Let $u : \mathbb{R} \to [0, \infty]$ and $\gamma : \mathbb{R} \to \mathbb{R}$ satisfy $\lim_{R\to\infty} \gamma(R) = +\infty$. Then*

$$\liminf_{R\to\infty} u(\gamma(R)) \ge \liminf_{R\to\infty} u(R).$$

Proof. Fix any threshold $X \in \mathbb{R}$ and set $m_X = \inf_{r \ge X} u(r)$. Since $\gamma(R) \to +\infty$ as $R \to \infty$, there exists $Y \in \mathbb{R}$ such that for all $R \ge Y$, $\gamma(R) \ge X$. Consequently, $u(\gamma(R)) \ge m_X$ for all $R \ge Y$, which implies $\inf_{R \ge Y} u(\gamma(R)) \ge m_X$, and hence $\liminf_{R\to\infty} u(\gamma(R)) \ge m_X$. Taking the supremum over all $X \in \mathbb{R}$ gives $\liminf_{R\to\infty} u(\gamma(R)) \ge \sup_X m_X = \liminf_{R\to\infty} u(R)$. □

Theorem 4.10 (Result D4: Open vs. closed disk equivalence)**.** *Let $C \subseteq \mathbb{R}^2$ be locally finite, $\rho > 0$, and let $g : (0, \infty) \to [0, \infty)$ be any function (no smoothness or monotonicity is needed). Then:*

(i) C has closed-disk density ρ ($\lim_{R\to\infty} \frac{N_R(C)}{\pi R^2} = \rho$) if and only if C has open-disk density ρ ($\lim_{R\to\infty} \frac{N^\circ_R(C)}{\pi R^2} = \rho$);

(ii) The open-disk and closed-disk lower energies coincide: $E^\circ_g(C) = E_g(C)$.

Proof. **(i) Density equivalence:** For every $R > 1$, the inclusions $\overline{B}(0, R-1) \subseteq B^\circ(0, R) \subseteq \overline{B}(0, R) \subseteq B^\circ(0, R+1)$ imply

$$\frac{N_{R-1}(C)}{\pi(R-1)^2} \cdot \frac{(R-1)^2}{R^2} \le \frac{N^\circ_R(C)}{\pi R^2} \le \frac{N_R(C)}{\pi R^2} \le \frac{N^\circ_{R+1}(C)}{\pi(R+1)^2} \cdot \frac{(R+1)^2}{R^2}.$$

Since $\frac{(R\pm1)^2}{R^2} \to 1$ as $R \to \infty$, the Squeeze Theorem (*Théorème des gendarmes*) immediately implies that $\frac{N_R(C)}{\pi R^2} \to \rho \iff \frac{N^\circ_R(C)}{\pi R^2} \to \rho$.
(ii) Lower energy equivalence: Because the disk energy depends on the radius R solely through the finite point set inside the disk, Lemma 4.8 gives, for every $R \in \mathbb{R}$,

$$E^\circ_g(C; R) = E_g(C; \varphi(R)) \qquad \text{and} \qquad E_g(C; R) = E^\circ_g(C; \psi(R)).$$

Since $\varphi(R) \ge R-1 \to +\infty$ and $\psi(R) \ge R \to +\infty$ as $R \to \infty$, applying Lemma 4.9 to both identities yields

$$E^\circ_g(C) = \liminf_{R\to\infty} E_g(C; \varphi(R)) \ge \liminf_{R\to\infty} E_g(C; R) = E_g(C),$$

and symmetrically $E_g(C) = \liminf_{R\to\infty} E^\circ_g(C; \psi(R)) \ge E^\circ_g(C)$. Thus $E^\circ_g(C) = E_g(C)$. □

Part V: Foundational Reference Appendix (*Math Sup / Math Spé* Refresher)

Reader's Guide: If your formal mathematical training dates back 30 years to the French **Math Sup / Math Spé** (*Classes Préparatoires aux Grandes Écoles*) curriculum of the 1990s, you already possess 90% of the algebraic and analytic machinery used in this document. The appendices below connect the classic theorems of *Math Sup / Math Spé* (Euclidean spaces, quadratic forms, $\liminf$ / $\limsup$, Riemann integrals, 1D Fourier series, and linear differential equations) to the modern concepts required for the universal optimality proof.

A Euclidean Spaces, Gram Matrices, and Dual Lattices

In *Math Spé* bilinear algebra, given a Euclidean space $(E, \langle\cdot,\cdot\rangle)$ of dimension $d = 2$ with orthonormal canonical basis (e_1, e_2) and any basis $\mathcal{B} = (v_1, v_2)$, let $P = \mathrm{Mat}_{(e_1,e_2)}(v_1, v_2)$ be the change-of-basis (2×2 transition) matrix whose columns are the coordinates of v_1 and v_2.

A.1 Covolume and Gram Matrix

1. **Covolume ($|\Lambda|$):** The area of the parallelogram spanned by v_1 and v_2 is the absolute value of the determinant:
$$|\Lambda| = |\det(v_1, v_2)| = |\det P|\,.$$
2. **Gram Matrix (G):** The matrix of the scalar product in the basis (v_1, v_2) is
$$G = \begin{pmatrix} \langle v_1, v_1\rangle & \langle v_1, v_2\rangle \\ \langle v_2, v_1\rangle & \langle v_2, v_2\rangle \end{pmatrix} = P^T P.$$
Notice that $\det G = \det(P^T P) = (\det P)^2 = |\Lambda|^2$. For any lattice vector $\lambda = jv_1 + kv_2 = P\binom{j}{k}$, its squared Euclidean norm is the positive-definite quadratic form
$$\|\lambda\|^2 = \left\langle P\begin{pmatrix} j \\ k\end{pmatrix}, P\begin{pmatrix} j \\ k\end{pmatrix}\right\rangle = \begin{pmatrix} j & k\end{pmatrix} P^T P \begin{pmatrix} j \\ k\end{pmatrix} = \begin{pmatrix} j & k\end{pmatrix} G \begin{pmatrix} j \\ k\end{pmatrix}.$$
For the triangular lattice A with $b = \frac{\sqrt{3}}{2}$, we have $P = \frac{1}{\sqrt{b}}\begin{pmatrix} 1 & 1/2 \\ 0 & b\end{pmatrix}$, so $\det P = \frac{1}{b}\cdot b = 1$ and $G = P^T P = \frac{1}{b}\begin{pmatrix} 1 & 1/2 \\ 1/2 & 1\end{pmatrix}$.

A.2 Dual Basis and Dual Lattice Λ^*

In linear algebra, the **dual basis** (v_1^*, v_2^*) in $\mathbb{R}^2$ (identifying $\mathbb{R}^2$ with its dual via the scalar product) is defined by the biorthogonality relation $\langle v_m^*, v_n\rangle = \delta_{mn}$ for $m, n \in \{1, 2\}$.

- In matrix form, if P^* is the matrix whose columns are v_1^*, v_2^*, the condition $(P^*)^T P = I_2$ means
$$P^* = \left(P^{-1}\right)^T = P\left(P^T P\right)^{-1} = PG^{-1}.$$
- Consequently, the Gram matrix of the dual lattice $\Lambda^* = \mathbb{Z}v_1^* + \mathbb{Z}v_2^*$ is $(P^*)^T P^* = G^{-1}$, and its covolume is $|\Lambda^*| = |\det P^*| = |\det P|^{-1} = 1/|\Lambda|$.

A.3 Asymptotic Lattice Point Counting

Lemma A.1 (Gauss's circle counting estimate for a lattice)**.** *Let $\Lambda \subset \mathbb{R}^2$ be a lattice of covolume $|\Lambda|$ and let $v \in \mathbb{R}^2$. Denote by $n(R) = \mathrm{card}((\Lambda + v) \cap \overline{B}(0, R))$. Denote by δ the diameter of the fundamental parallelogram $D = \{t_1 v_1 + t_2 v_2 : t_1, t_2 \in [-1/2, 1/2)\}$. Then for all $R \geq \delta$,*

$$\frac{\pi(R-\delta)^2}{|\Lambda|} \leq n(R) \leq \frac{\pi(R+\delta)^2}{|\Lambda|}, \qquad \textit{so} \qquad n(R) = \frac{\pi R^2}{|\Lambda|} + O(R).$$

Proof. The disjoint translates $\{x + D : x \in \Lambda + v\}$ tile $\mathbb{R}^2$, and each tile has area $\mathrm{Area}(D) = |\Lambda|$. Since $0 \in D$, every $d \in D$ satisfies $\|d\| \leq \delta$. For every $x \in (\Lambda + v) \cap \overline{B}(0, R)$, the triangle inequality shows $x + D \subseteq \overline{B}(0, R+\delta)$; conversely, the disk $\overline{B}(0, R-\delta)$ is completely covered by those tiles $x + D$ with $x \in (\Lambda + v) \cap \overline{B}(0, R)$. Comparing the total areas yields $\pi(R-\delta)^2 \leq n(R)\,|\Lambda| \leq \pi(R+\delta)^2$. □

B Limit Inferior (lim inf), Limit Superior (lim sup), and $[0, \infty]$

In *Math Sup* analysis, a bounded real sequence (u_n) can oscillate without converging, and is bounded by its lower and upper asymptotic envelopes.

B.1 Definition Over a Continuous Parameter $R \to \infty$

For a function $F : (0, \infty) \to [0, \infty]$ (such as the disk energy $F(R) = E_g(C; R)$), define for each threshold $R_0 > 0$ the infimum and supremum over the tail $[R_0, \infty)$:

$$I(R_0) := \inf_{R \geq R_0} F(R), \qquad S(R_0) := \sup_{R \geq R_0} F(R).$$

As the starting radius R_0 increases, the set $[R_0, \infty)$ shrinks. Therefore:

- $R_0 \mapsto I(R_0)$ is **monotone increasing**, so it always has a limit in $[0, \infty]$ as $R_0 \to \infty$ (namely its supremum):

$$\liminf_{R \to \infty} F(R) := \lim_{R_0 \to \infty} \left(\inf_{R \geq R_0} F(R) \right) = \sup_{R_0 > 0} \inf_{R \geq R_0} F(R).$$

- $R_0 \mapsto S(R_0)$ is **monotone decreasing**, so it always has a limit in $[0, \infty]$ as $R_0 \to \infty$ (namely its infimum):

$$\limsup_{R \to \infty} F(R) := \lim_{R_0 \to \infty} \left(\sup_{R \geq R_0} F(R) \right) = \inf_{R_0 > 0} \sup_{R \geq R_0} F(R).$$

We always have $\liminf_{R\to\infty} F(R) \leq \limsup_{R\to\infty} F(R)$, with equality if and only if $\lim_{R\to\infty} F(R)$ exists in $[0, \infty]$.

B.2 Why lim inf Is Super-Additive (and Why Equality Fails)

A crucial point in Section 3.2 is that lim inf is **super-additive**, not linear: for any two non-negative functions u, w,

$$\liminf_{R \to \infty} (u(R) + w(R)) \geq \liminf_{R \to \infty} u(R) + \liminf_{R \to \infty} w(R). \tag{10}$$

Proof. For every $R \geq R_0$, $u(R) + w(R) \geq \inf_{r \geq R_0} u(r) + \inf_{r \geq R_0} w(r)$. Taking the infimum over $R \geq R_0$ on the left and letting $R_0 \to \infty$ gives (10). □

Counterexample to equality: Take $u_n = 1 + (-1)^n$ (which alternates $2, 0, 2, 0, \ldots$) and $w_n = 1 - (-1)^n$ (which alternates $0, 2, 0, 2, \ldots$). Then $\liminf u_n = 0$ and $\liminf w_n = 0$, whereas $u_n + w_n = 2$ for all n, so $\liminf(u_n + w_n) = 2 > 0 + 0$. Notice that the inequality (10) goes in the **exact direction we need** to bound $E_g(C)$ from below.

C From *Math Spé* Riemann Integrals to Measure Theory, Fatou, and Tonelli

In the 1990s *Math Spé* syllabus, integration was taught via regulated/piecewise-continuous functions on intervals, and interchanging $\int$ and $\sum$ required proving uniform convergence or bounding by a fixed integrable dominating function (*Théorème de convergence dominée*). In Lebesgue measure theory, when the functions involved are **non-negative** (≥ 0), two much more flexible theorems hold *without* needing uniform convergence or an upper dominating bound.

C.1 Borel Measures μ on $[0,\infty)$: Unifying Sums and Densities

When Bernstein's theorem writes $g(t) = \int_{[0,\infty)} e^{-\pi\alpha t}\, d\mu(\alpha)$, the Borel measure μ simply generalizes two familiar *Math Spé* cases into a single notation:

- **Discrete weighted sum (Dirac masses):** If $\mu = \sum_{k=1}^{m} c_k \delta_{\alpha_k}$ with $c_k \geq 0$, then $\int_{[0,\infty)} \phi(\alpha)\, d\mu(\alpha) = \sum_{k=1}^{m} c_k \phi(\alpha_k)$.
- **Continuous density:** If $d\mu(\alpha) = w(\alpha)\, d\alpha$ for a continuous weight $w(\alpha) \geq 0$ (such as $w(\alpha) = \frac{\pi^{s/2}}{\Gamma(s/2)} \alpha^{s/2-1}$ for Riesz potentials), then $\int_{[0,\infty)} \phi(\alpha)\, d\mu(\alpha) = \int_0^\infty \phi(\alpha) w(\alpha)\, d\alpha$ is an ordinary improper Riemann integral.

C.2 Fatou's Lemma and Tonelli's Theorem

Theorem C.1 (Fatou's Lemma)**.** *Let $(f_n)_{n\in\mathbb{N}}$ be any sequence of non-negative measurable functions $f_n : [0,\infty) \to [0,\infty]$. Then*

$$\int_{[0,\infty)} \left(\liminf_{n\to\infty} f_n(\alpha) \right) d\mu(\alpha) \leq \liminf_{n\to\infty} \int_{[0,\infty)} f_n(\alpha)\, d\mu(\alpha).$$

Intuition: Fatou's Lemma is simply the continuous-integral version of the super-additivity of lim inf in (10). Because $f_n \geq 0$, no hypothesis of uniform convergence or domination is required; the inequality holds in $[0,\infty]$ unconditionally.

Theorem C.2 (Tonelli's Theorem for Series and Integrals of Non-Negative Terms)**.** *Let $(u_k(\alpha))_{k\in\mathbb{N}}$ be a sequence of non-negative measurable functions $u_k : [0,\infty) \to [0,\infty]$. Then*

$$\int_{[0,\infty)} \left(\sum_{k=1}^{\infty} u_k(\alpha) \right) d\mu(\alpha) = \sum_{k=1}^{\infty} \int_{[0,\infty)} u_k(\alpha)\, d\mu(\alpha) \qquad \text{in } [0,\infty].$$

Intuition: In *Math Spé*, to swap $\sum$ and $\int$, one had to check beforehand that the series of integrals converged ($\sum \int |u_k| < \infty$, Fubini's theorem). When $u_k(\alpha) \geq 0$, Tonelli's theorem says the two sides are **always equal in** $[0,\infty]$: either both are finite and equal to the same real number, or both are $+\infty$.

D Completely Monotone Functions, Leibniz's Rule, and Gamma Integrals

D.1 Closure Properties of Completely Monotone Functions

Recall from Definition 1.8 that $g : (0,\infty) \to \mathbb{R}$ is completely monotone if $g \in C^\infty(0,\infty)$ and $(-1)^r g^{(r)}(t) \geq 0$ for all $r \geq 0$ and $t > 0$.

Proposition D.1 (Product and composition closure)**.** *(i) If f and g are completely monotone on $(0,\infty)$, then their product $f \cdot g$ is completely monotone.*

(ii) If f is completely monotone on $(0,\infty)$ and $h : (0,\infty) \to (0,\infty)$ is C^∞ with a completely monotone first derivative h', then $f \circ h$ is completely monotone.

Proof. (i) By the *Math Sup* Leibniz rule for the r-th derivative of a product,

$$(-1)^r (fg)^{(r)}(t) = \sum_{k=0}^{r} \binom{r}{k} \left[(-1)^k f^{(k)}(t)\right] \left[(-1)^{r-k} g^{(r-k)}(t)\right] \geq 0,$$

since every term in the sum is a product of non-negative quantities.
(ii) For $r = 1$, $-(f\circ h)'(t) = [-f'(h(t))]\cdot h'(t) \geq 0$. By induction on r and part (i), every higher derivative $(-1)^r (f \circ h)^{(r)}$ is a positive linear combination of products of $(-1)^k f^{(k)} \circ h$ and $(-1)^m (h')^{(m)}$, all of which are non-negative. □

Example D.2 (Why Riesz and Yukawa potentials are completely monotone)**.**

- **Riesz ($g_s(t) = t^{-s/2}$, $s > 0$):** Using the Euler Gamma function $\Gamma(\sigma) = \int_0^\infty u^{\sigma-1} e^{-u}\, du$ ($\sigma = s/2 > 0$), the change of variable $u = \pi\alpha t$ gives the explicit Bernstein representation:

$$t^{-s/2} = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^\infty e^{-\pi\alpha t}\, \alpha^{s/2-1}\, d\alpha.$$

- **Yukawa ($V(r) = e^{-\kappa r}/r \implies g(t) = t^{-1/2} e^{-\kappa\sqrt{t}}$, $\kappa > 0$):** Here $t^{-1/2}$ is completely monotone ($s = 1$), and $h(t) = \kappa\sqrt{t} > 0$ has derivative $h'(t) = \frac{\kappa}{2} t^{-1/2}$, which is completely monotone. Since $u \mapsto e^{-u}$ is completely monotone, $e^{-\kappa\sqrt{t}}$ is completely monotone by (ii), and the product $t^{-1/2} e^{-\kappa\sqrt{t}}$ is completely monotone by (i).

E 2D Fourier Analysis, Gaussian Self-Duality, and Poisson Summation

E.1 Fourier Transform of a Gaussian via a *Math Spé* Differential Equation

Proposition E.1 (Fourier transform of G_α)**.** *For any $\alpha > 0$ and $G_\alpha(x) = e^{-\pi\alpha\|x\|^2}$ on $\mathbb{R}^2$, we have*

$$\widehat{G_\alpha}(\xi) = \int_{\mathbb{R}^2} e^{-\pi\alpha\|x\|^2} e^{-2\pi i\langle x,\xi\rangle}\, dx = \frac{1}{\alpha} e^{-\pi\|\xi\|^2/\alpha}.$$

Proof. By Fubini's theorem, the 2D integral factors into the product of two identical 1D integrals: $\widehat{G_\alpha}(\xi_1,\xi_2) = \phi(\xi_1)\phi(\xi_2)$, where $\phi(\xi) = \int_{-\infty}^{\infty} e^{-\pi\alpha x^2} e^{-2\pi i x\xi}\, dx$. Differentiating under the integral sign with respect to ξ and integrating by parts gives

$$\phi'(\xi) = \int_{-\infty}^{\infty} (-2\pi i x) e^{-\pi\alpha x^2} e^{-2\pi i x\xi}\, dx = \frac{i}{\alpha} \int_{-\infty}^{\infty} \frac{d}{dx}\left(e^{-\pi\alpha x^2}\right) e^{-2\pi i x\xi}\, dx = -\frac{2\pi\xi}{\alpha}\phi(\xi).$$

Solving this first-order linear homogeneous ODE gives $\phi(\xi) = \phi(0) e^{-\pi\xi^2/\alpha}$. Since $\phi(0) = \int_{-\infty}^{\infty} e^{-\pi\alpha x^2}\, dx = \frac{1}{\sqrt{\alpha}}$, multiplying $\phi(\xi_1)\phi(\xi_2)$ yields $\frac{1}{\alpha} e^{-\pi(\xi_1^2+\xi_2^2)/\alpha}$. □

E.2 The 2D Poisson Summation Formula from *Math Spé* Fourier Series

Theorem E.2 (2D Poisson Summation Formula)**.** *Let $\Lambda \subset \mathbb{R}^2$ be a lattice of covolume $|\Lambda|$ and let $f \in \mathcal{S}(\mathbb{R}^2)$ be a Schwartz function (smooth and decaying faster than any inverse power of $\|x\|$ along with all its derivatives). For every $v \in \mathbb{R}^2$,*

$$\sum_{\lambda \in \Lambda} f(\lambda + v) = \frac{1}{|\Lambda|} \sum_{w \in \Lambda^*} \widehat{f}(w) e^{2\pi i \langle w, v \rangle}.$$

Proof. Consider the periodized function $F(v) = \sum_{\lambda \in \Lambda} f(\lambda + v)$. Because $f \in \mathcal{S}(\mathbb{R}^2)$, the series and all its term-by-term partial derivatives converge normally on compact subsets of $\mathbb{R}^2$, so F is C^∞ and Λ-periodic ($F(v + \lambda) = F(v)$ for all $\lambda \in \Lambda$). Just as a 1D T-periodic function is expanded in harmonics $e^{2\pi i n x/T}$, a Λ-periodic function on $\mathbb{R}^2/\Lambda$ is expanded in the orthogonal basis of characters $\{v \mapsto e^{2\pi i \langle w, v \rangle} : w \in \Lambda^*\}$. Its Fourier coefficient at frequency $w \in \Lambda^*$ over a fundamental domain D of area $|\Lambda|$ is

$$\begin{aligned} c_w(F) &= \frac{1}{|\Lambda|} \int_D F(v) e^{-2\pi i \langle w, v \rangle} \, dv = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} \int_D f(\lambda + v) e^{-2\pi i \langle w, \lambda + v \rangle} \, dv \\ &= \frac{1}{|\Lambda|} \int_{\mathbb{R}^2} f(x) e^{-2\pi i \langle w, x \rangle} \, dx = \frac{\widehat{f}(w)}{|\Lambda|}, \end{aligned}$$

where we used $\langle w, \lambda \rangle \in \mathbb{Z} \implies e^{-2\pi i \langle w, \lambda \rangle} = 1$ and the tiling $\mathbb{R}^2 = \bigsqcup_{\lambda \in \Lambda} (\lambda + D)$. Since F is C^∞, its Fourier series converges absolutely to $F(v)$ for every $v \in \mathbb{R}^2$, yielding the formula. □

F Double Zeros and Sine-Product Cardinal Interpolation

Why does the 2D auxiliary function f_α (as described by upstream) use a **sine product with double zeros**?

F.1 Why Minorants Must Have Double Zeros at Contact Points

Proposition F.1 (First-order tangency at lattice shells)**.** *Suppose $f_\alpha \in \mathcal{S}(\mathbb{R}^2)$ satisfies $f_\alpha(x) \leq G_\alpha(x)$ for all $x \neq 0$ and $f_\alpha(a) = G_\alpha(a)$ at a lattice vector $a \in A \setminus \{0\}$. Then*

$$\nabla f_\alpha(a) = \nabla G_\alpha(a).$$

Proof. The smooth difference function $h(x) = G_\alpha(x) - f_\alpha(x)$ satisfies $h(x) \geq 0$ on $\mathbb{R}^2 \setminus \{0\}$ and $h(a) = 0$. Therefore a is an interior local minimum of h on the open set $\mathbb{R}^2 \setminus \{0\}$. By Fermat's theorem on stationary points (*Math Sup* calculus), $\nabla h(a) = 0$. In terms of the radial variable $t = \|x\|^2$, if $G_\alpha(x) - f_\alpha(x) = \Delta(\|x\|^2)$, then $\Delta(t_m) = 0$ and $\Delta'(t_m) = 0$ at every nonzero norm shell $t_m = \|a\|^2$: every contact shell must be a **root of even multiplicity** ≥ 2. □

F.2 From Euler's Sine Product to Cardinal Interpolation with Double Zeros

In classical analysis, Euler's infinite product for the sine function is

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right),$$

which has simple zeros at every integer $z \in \mathbb{Z}$. Squaring this function gives $\sin^2(\pi z)$, which has **double zeros** at every integer $z = n \in \mathbb{Z}$ and remains ≥ 0 for all real $z \in \mathbb{R}$.

- If we divide $\sin^2(\pi z)$ by $\pi^2(z-n)^2$, we obtain the Fejér kernel $\left(\frac{\sin(\pi(z-n))}{\pi(z-n)}\right)^2$, which equals 1 with derivative 0 at $z=n$, and equals 0 with derivative 0 at every other integer $m \neq n$.
- Multiplying $\sin^2(\pi z)$ by simple-pole factors $\frac{1}{z-n}$ and double-pole factors $\frac{1}{(z-n)^2}$ allows one to construct an entire function (Hermite–Fejér cardinal interpolant) that takes **prescribed values and prescribed first derivatives** at any finite reference list of nodes, while maintaining double zeros and a controlled sign at all remaining nodes.

In the upstream 2D argument, because the triangular lattice shells $n = j^2+jk+k^2$ form a subset of the integers $\mathbb{N}$, this sine-product cardinal interpolation (combined with Gaussian damping and certified finite quadrature) is used in place of the modular forms of dimensions 8 and 24. As noted in Section 2.4, the product also places double zeros at the non-Löschian integers; whether and how the upstream construction accounts for this is one of the points a reviewer of the upstream manuscript should check.

G Printable Pen-and-Paper Review Checklist

References

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[OAI26] `openai/math` repository, theorem `OAI.AtomicTriangular.universal_energy_minimum`, commit `adc7f124` (October 2026). AI-generated proof and manuscript; Lean-kernel-checked against the release's own definitions; not peer reviewed.

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[Wid41] D. V. Widder, *The Laplace Transform*, Princeton University Press, 1941.

[XC26] `xaviercallens/triangular-universal-optimality-lean`, *Conditional corollaries of an AI-generated `openai/math` theorem, not a proof of the conjecture*, GitHub (2026), https://github.com/xaviercallens/triangular-universal-optimality-lean. Zenodo: `10.5281/zenodo.23285631` (D1, D1b, D2) and `10.5281/zenodo.23293429` (D3).

Section	Mathematical Claim	Key Mechanism to Verify	Reviewer Sign-off
Prop. 1.5 & 1.7	$\lvert A\rvert = 1$, $\lVert p(j,k)\rVert^2 = \frac{2}{\sqrt{3}}(j^2+jk+k^2)$, and $A^* = JA$.	Matrix determinant $\det(v_1, v_2) = 1$ and biorthogonality $\langle Jv_m, v_n\rangle \in \mathbb{Z}$.	☐ Verified
Thm. 2.3	Periodic LP bound $E_{g_\alpha}^{\mathsf{per}}(C) \geq \widehat{f_\alpha}(0) - f_\alpha(0) = \mathcal{E}_{g_\alpha}(A)$.	Minorant $f_\alpha \leq G_\alpha$, diagonal $Nf_\alpha(0)$, 2D Poisson summation, $\widehat{f_\alpha} \geq 0$.	☐ Verified
Prop. 2.4	Periodic disk energy converges: $\lim_{R\to\infty} E_{g_\alpha}(C;R) = E_{g_\alpha}^{\mathsf{per}}(C)$.	Annulus cut C_{R-L} and Gaussian tail bound $\tau(L) \to 0$.	☐ Verified
Prop. 2.5	LP bound for *arbitrary* locally finite C (cited).	Positive-definiteness applied to points minus uniform background; check against [CdCI18, Prop. 2.2].	☐ Checked against source
Claim 2.6	Sharp 2D f_α exists for every $\alpha > 0$ (**imported, unverified**).	Upstream sine-product construction and ball-arithmetic certificates; *cannot* be verified from this document.	☐ Upstream audit pending
Thm. 1.10	Passage from Gaussians G_α to all completely monotone g.	Fatou's Lemma along (R_n), atom at $\alpha = 0$ (`energy_one`), Tonelli's Theorem.	☐ Verified
Result D1 & D1b	Riesz r^{-s} admissibility ($s > 0$) and finiteness ($\iff$ $s > 2$).	Pochhammer identity $(-\sigma)_r^{\downarrow} = (-1)^r(\sigma)_r^{\uparrow}$ and shell bound $\sum 8n \cdot n^{-s}$.	☐ Verified
Result D2 (Thm. 4.5)	Optimality of $\rho^{-1/2}A$ at arbitrary density $\rho > 0$.	Bijection $(cC)_R = c(C_{R/c})$, energy scaling $E_h(cC) = E_{h(c^2\cdot)}(C)$.	☐ Verified
Result D3 (Thm. 4.7)	Vacancy $V = A \setminus \{0\}$ is optimal but not isometric to A.	Factor $\kappa(R) = \frac{N}{N-1} \to 1$ and affine midpoint reflection $f(2a-b) = 2f(a) - f(b)$.	☐ Verified
Result D4 (Thm. 4.10)	Open disks $B^\circ(0,R)$ and closed disks $\overline{B}(0,R)$ coincide.	Finite set radii $\varphi(R) \in [R-1, R)$ and $\psi(R) \in (R, R+1]$ + Squeeze Theorem.	☐ Verified

Table 1: Pen-and-paper review checklist for auditing the human proofs alongside the Lean 4 formalization.